

MSAF: A Multi-Level Sensing Adversarial Framework for Signal Recovery in Synthetic Aperture Ultrasound

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Abstract—Synthetic aperture ultrasound imaging achieves two-way focusing and produces high-quality image formation by coherently combining returning echoes from successive single-element or sub-aperture transmissions across the entire transducer array. To depict transient or fast tissue dynamics, a prominent line of research efforts involves employing a limited number of sub-apertures to approximate full-aperture scans, enabling full-view imaging at high frame rates. However, existing end-to-end approaches predominantly operate in the image domain, which inherently discards phase information. Some studies use radio-frequency or in-phase and quadrature (IQ) domains, but they analyze spatial features frame by frame, thus not fully leveraging the temporal information. Motivated by high-dimensional data inherent to synthetic apertures, this study proposes a multi-level sensing adversarial framework (MSAF) to recover full IQ signals from sparse sub-aperture transmissions. MSAF utilizes sparse apertures by spatiotemporal sensing and constrains the recovered signals by global-attribute sensing. MSAF was evaluated on large in vivo human echocardiography datasets and on in vivo unseen echocardiographic views and abdominal images as zero-shot testing sets. Experimental results demonstrated high-fidelity IQ signal recovery, achieving a peak signal-to-noise ratio of 32.80 dB, a structural similarity index of 0.88, a contrast-to-noise ratio of 2.92 dB, and a contrast ratio of 15.46 dB, outperforming all tailored benchmark models. Therefore, MSAF enables sparse-aperture transmissions with high signal quality and data efficiency, making it a potential approach for functional ultrasound imaging.

Index Terms—Adversarial learning, in-phase and quadrature signal, radio-frequency, synthetic aperture, ultrafast ultrasound.

I. INTRODUCTION

SYNTHETIC aperture ultrasound for medical applications emerged in the 1990s to address the issues of single transmit focus and frame rate in conventional diagnostic ultrasound imaging [1]. It synthesizes transmit focusing by combining beam-formed images from individual single-element or multi-element transmissions [1]. This approach therefore achieves two-way focusing everywhere in the field of view and improves contrast and spatial resolution. About two decades later, synthetic aperture ultrasound with multi-element sub-aperture transmissions of diverging waves was demonstrated to produce high-quality ultrasound images of the heart at thousands of frames per second [2]. The echo signals received by each sub-aperture are beamformed as a low-quality image. The low-quality images from all sub-apertures are then coherently compounded to generate a high-quality image, which is dynamically focused in transmission and in reception [2]. Currently, synthetic aperture ultrasound has become a prevalent in vivo ultrafast ultrasound imaging method to acquire dynamic biomedical information.

More sub-apertures can enhance spatial resolution and contrast but at the expense of frame rates (temporal resolution). Balancing image quality and frame rate is essential for reliable detection and quantification of transient tissue dynamics [3]. For example, electromechanical wave imaging [4] necessitates sufficient quality image data at a frame rate higher than 500 frames/second to pinpoint a cardiac conduction block and aortic closure wave [5]; Myocardial stiffness can be assessed by shear wave elastography, which also relies on ultrafast ultrasound to capture remotely generated fast-propagating shear waves through acoustic radiation force [6]. The inherent trade-off between image quality and frame rate limits the diagnostic capabilities of ultrafast ultrasound with synthetic transmit aperture in echocardiography to reveal unprecedented inherent physiological events in the heart [2], [7].

The aforementioned clinical applications require not only B-mode imaging for real-time visual feedback but also access to raw signals, i.e., in-phase and quadrature (IQ) signals or radio-frequency (RF) signals, for tasks like tissue motion estimation.

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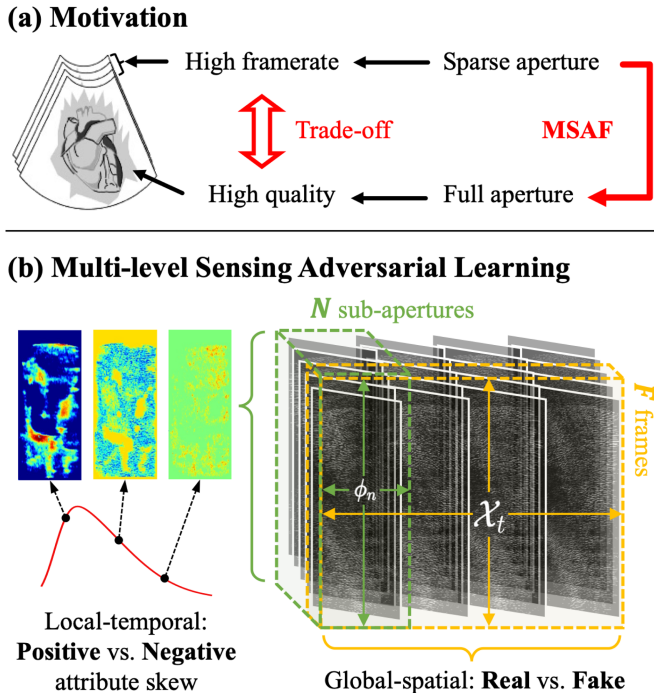


Fig. 1. (a) This work solves the trade-off between high frame rate and high quality in synthetic aperture ultrasound by recovering beamformed in-phase and quadrature signals from sparse sub-aperture transmissions; (b) The proposed multi-level sensing adversarial framework (MSAF) to recover high-dimensional signals.

Compared with raw signals, B-mode images lose critical phase information during envelope detection, which is followed by log compression. In contrast to RF signals, IQ data have a smaller data size, enabling efficient data transfer and storage. Hence, this study focuses on recovering IQ signals in synthetic aperture ultrasound with sparse transmissions (Fig. 1(a)).

Signal recovery in synthetic aperture ultrasound presents two challenges: spatiotemporal information preservation and high-dimensional signal processing. Unlike natural images, synthetic aperture ultrasound has rich high-dimensional signals. As shown in Fig. 1(b), one single set of synthetic aperture ultrasound data comprises a spatial series of sub-aperture data (N sub-apertures in total; green dashed box), with each sub-aperture dataset containing a temporal sequence of frames over the total acquisition time (F frames in total; yellow dashed box). At a given time point, each sub-aperture yields a low-resolution image frame, and standard synthetic aperture ultrasound produces a full-aperture frame by coherently compounding N low-resolution frames. The way to handle these high-dimensional signals in neural networks is yet to be explored for the efficient recovery of IQ signals in synthetic aperture ultrasound.

Most related works adopt a frame-by-frame recovery scheme, which aligns with the common method used for natural images. More specifically, for dataset construction, prior research concatenated echo signals from different sub-apertures along the tensor's channel direction [8]. This method regards multiple sub-apertures as distinct feature channels of a neural network. It allows the model to capture spatial features within individual frames, but does not account for the dynamic features over time. Thus, the preservation of temporal correlations across different

frames pertaining to a sub-aperture is limited. Furthermore, feature channels in CNNs are treated as unordered [9], which disrupts the spatial relationships among multiple sub-apertures. A method that senses and recovers both spatial and temporal information and models the spatio-temporal relationships of signals between sub-apertures is imperative.

To address the aforementioned challenges, this paper proposes a multi-level sensing adversarial framework (MSAF) for recovering beamformed IQ signals in synthetic aperture ultrasound. The study objective is to restore a full-aperture dataset from a limited number of sub-aperture transmission events, thereby improving image quality at high frame rates. Additionally, the use of sparse sub-apertures can significantly reduce system complexity in ultrafast ultrasound, while enabling adaptive beamforming and flexible sub-aperture selection for compounding. Based on an adversarial learning paradigm, MSAF features the following contributions:

- *Aperture-wise Attribute Discrimination*: This component constrains the local fidelity of recovered signals from each sub-aperture over a time period;
- *Frame-wise Global Discrimination*: This component constrains the global fidelity of each full-aperture frame, which is compounded coherently from all sub-apertures;
- *Complex Spatiotemporal Generator*: This component integrates a complex-valued U-Net and a complex-valued LSTM to capture spatiotemporal information in beamformed IQ sequences.
- *High-quality Signal Recovery*: On average, MSAF attains 32.80 dB PSNR and 0.88 SSIM, outperforming benchmark models by 13.28% and 7.62%, respectively.

The multi-level sensing in MSAF encompasses spatial, temporal, global, and attribute (local) levels, specifically designed for synthetic aperture ultrasound signals.

II. RELATED WORKS

A. Conventional Ultrasound Image Enhancement

Previous efforts to balance image quality and frame rates in synthetic aperture ultrasound can be broadly categorized into signal processing and deep learning approaches. Signal processing methods primarily focus on signal recovery through compressed sensing and tensor completion, as well as image quality enhancement via adaptive beamforming [10] and coded excitation [11]. Zhang et al. [12] accelerated imaging through compressed-sensing in the IQ domain by a minimal L_2 -norm least squares method. Afrakhteh et al. [13] utilized tensor completion to reconstruct images from random synthetic transmit apertures. Anand et al. [14] implemented a genetic algorithm to select receive elements, thereby optimizing the contrast-to-noise ratio (CNR) of compressed sensing synthetic aperture ultrasound. However, conventional methods rely on time-consuming iterative optimization and can be susceptible to in vivo scenarios with low-quality signals. The integration of deep learning can improve their performance. For example, Zhang et al. [15] proposed a model to estimate the null space component of compressed-sensing synthetic aperture ultrasound from fewer transmissions.

B. Deep Learning-Based Ultrasound Image Reconstruction

Deep learning methods establish end-to-end mappings between low-quality signals and targeted ones in a data-driven manner, leveraging neural networks' robust nonlinear modeling capability [16]. Neural networks for the recovery of ultrasound signals from incomplete data to reconstruct high-quality images can be categorized by operational domains into image (i.e., envelope data) and raw signal (RF or IQ) domains.

1) *Image Domain*: This class of models utilizes end-to-end networks that learn information solely in the B-mode image domain and directly output a B-mode image. It has been demonstrated for coherent plane wave compounding with a limited number of steered plane wave transmissions [17], [18], [19]. These studies have shown that the model-reconstructed B-mode image from a few (e.g., 1 or 3) transmission angles can exhibit spatial resolution and contrast comparable to the reference B-mode image, which is coherently compounded from a large number (e.g., 75 or 31) of steered plane wave transmissions. However, the model output does not contain phase information, making it insufficient for applications such as elasticity imaging.

2) *Raw Signal Domain*: In the RF domain, Yoon et al. [8] exploited redundancy in the receiver-transmit plane to interpolate missing RF signals. Chen et al. [20] proposed RxNet to adaptively select optimized receiver elements and recover signals of inactive elements. Zhang et al. [21] recovered RF signals of compressed sensing synthetic aperture ultrasound from partial Hadamard-encoded plane waves. In the IQ domain, Senou et al. [22] trained a U-Net using pre-beamformed IQ signals to recover single-line acquisition from multi-line acquisitions. Vedula et al. [23] recovered single-line transmission from multi-line transmission. Both multi-line acquisition and transmission methods highlight the versatility of CNNs in addressing different aspects of ultrasound data. As for beamformed IQ signals, Khan et al. [24] constructed neural networks as beamformers to generate beamformed IQ signals from sub-sampled channel signals.

Aside from ultrasound signal recovery, Peretz et al. [25] employed a deep neural network as a beamformer to transfer data from a small aperture into a larger one, thus improving resolution and contrast with few elements in an ultrasound array; Chen et al. [26] introduced ApodNet, which utilizes trained binary weights for apodization in plane wave transmissions; Edoardo et al. [27] applied a U-Net to transform monostatic synthetic aperture ultrasound images into multi-static representations. Different operational domains offer complementary information [15]. However, to the best of our knowledge, no prior studies have concurrently incorporated multiple domains into the signal recovery task in synthetic aperture ultrasound.

C. Generative Adversarial Networks

Generative Adversarial Networks (GANs) have emerged as a state-of-the-art paradigm for medical image restoration [28]. Comprising a generator and a discriminator engaged in an adversarial learning process, GANs learn to map low-quality images

with noise, artifacts, or low resolution to their high-fidelity counterparts.

A general GAN quantifies the difference between the real and generated data via a single scalar output by the discriminator [29]. The optimization of GAN is a min-max game:

$$\min_{\mathbf{G}} \max_{\mathbf{D}} \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log \mathbf{D}(\mathbf{x}) - 0] + \mathbb{E}_{\mathbf{x} \sim p_{\mathbf{G}}} [\log(1 - \mathbf{D}(\mathbf{x}))], \quad (1)$$

where $\mathbf{G}$ is a generator and $\mathbf{D}$ is a discriminator.

However, as shown in Fig. 1(b), the high-dimensional signal of synthetic aperture ultrasound features an additional temporal dimension F and an aperture dimension N . Therefore, a generator for a single natural image or a discriminator that performs binary classification cannot simultaneously ensure both spatial and temporal fidelity, nor can it accommodate both global structure and local details of the IQ signals [30].

III. METHODOLOGY

A. Problem Statement

Let $\mathbf{y} \in \mathbb{C}^{F \times N \times H \times W}$ represent the beamformed IQ signals from sparse-aperture transmissions with N sub-apertures and F frames. $H \times W$ corresponds to the field of view. Let $\mathbf{x} \in \mathbb{C}^{F \times M \times H \times W}$ ($N < M$) denote the full-aperture signals from M sub-apertures. The relationship between $\mathbf{x}$ and $\mathbf{y}$ is modeled as $\mathbf{y} = \mathbf{F}_u \mathbf{x} + \epsilon$, where $\mathbf{F}_u$ is an aperture-wise under-sampling operator and ϵ is electronic noise [31]. The under-determined inversion function for estimating the missing signals, $\tilde{\mathbf{x}} \in \mathbb{C}^{F \times (M-N) \times H \times W}$, is expressed as $\tilde{\mathbf{x}} = \mathbf{F}_u^{-1}(\mathbf{y} - \epsilon) - \mathbf{y}$.

MSAF is designed to solve this inverse problem of recovering the beamformed IQ signals corresponding to the missing sub-apertures from available sparse-aperture transmissions. This study uses equally spaced interval sampling to implement sparse transmissions, and the undersampling factor, $\mathbf{F}_u$, is 2 so that $M = 2 \times N$ (Fig. 2(a)). The generator recovers missing sub-aperture signals ($\tilde{\mathbf{x}}$) using signals from active sparse sub-apertures ($\mathbf{y}$). The discriminator then constrains signal quality at both global-spatial and temporal-attribute levels.

An image can be considered as a collection of multiple visual attributes [32]. Each spatial attribute represents a local region of a semantic concept. Similarly, we regard the IQ signals from a single sub-aperture over a time period as a collection of temporal attributes. Given ϕ_n , the IQ signal from the n -th sub-aperture, the discriminator extracts its attribute set, denoted as $\mathcal{A}_n \subset \sum_t \phi_n$. We propose aperture-wise attribute discrimination to ensure that the attributes of the recovered signals closely approximate those of the real signals in each sub-aperture (Method III.B.2). Given the t -th frame, the discriminator also extracts global spatial feature, denoted as $\mathbf{X}_t$; a frame-wise global discrimination is thus proposed to approximate the recovered signals of each frame to the real ones (Method III.B.1).

Therefore, the signal recovery is formulated as a new minimization problem as follows:

$$\min_{\mathbf{x}} \sum_n \left\| \mathcal{A}_n - \tilde{\mathcal{A}}_n \right\| + \sum_t \left\| \mathbf{X}_t - \tilde{\mathbf{X}}_t \right\|$$

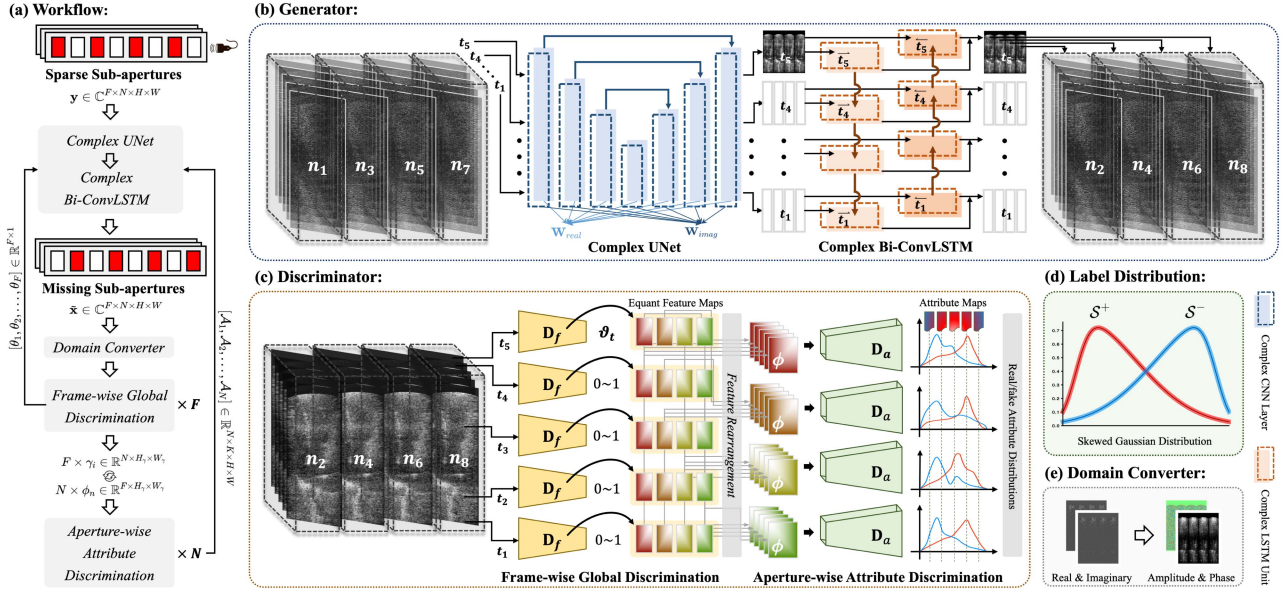


Fig. 2. (a) Workflow of the proposed MSAF. (b) The generator cascades a complex UNet and a complex Bi-convLSTM. (c) The discriminator begins with frame-wise global discrimination, followed by aperture-wise attribute discrimination. D_a decodes K attribute maps and formulates an attribute distribution for each sub-aperture. (d) Two distributions labeling fake and real. (e) The domain converter linking the generator and discriminator.

$$+ \|\mathbf{x} - \tilde{\mathbf{x}}\|_2^2 + \mathcal{R}(\mathbf{x}), \quad (2)$$

where symbols with a tilde are from recovered signals, and those without a tilde are from targeted ones. The first term represents aperture-wise attribute discrimination, which constrains the local attributes within a single sub-aperture across consecutive frames; the second term represents frame-wise global discrimination, which constrains the global spatial fidelity within a frame (coherently compounded from all sub-apertures); the third term is a per-pixel data error term optimized using a Complex Spatiotemporal Generator (Method III.C); the last term denotes an L_2 weight regularization.

B. Multi-Level Sensing Discriminator

A multi-level sensing discriminator is proposed for adversarial learning on spatial and temporal, global and attribute (local) levels. It is a two-stage network that performs frame-wise global discrimination (D_f) and aperture-wise attribute discrimination (D_a) sequentially (Fig. 2(c)).

1) **Frame-Wise Global Discrimination:** In the first stage, D_f independently discriminates each frame based on its intraframe global features. D_f consists of three 2D convolutional layers and three fully connected layers. The convolutional layers have a kernel size of 3×3 , and channel numbers of 32, 64, and 128, respectively. Max pooling reduces the feature size by $\frac{1}{2}$ after each convolution. For the t -th frame, its amplitude and phase are stacked as a two-channel input. The convolutional layers extract intraframe spatial features, yielding feature maps γ_t . Then, the fully connected and sigmoid layers predict a probability $\theta_t \in [0, 1]$, where 0 means fake and 1 means real.

Hence, the objective function of $\mathcal{D}_f$ is

$$\min_{\mathbf{G}} \max_{\mathbf{D}_f} \sum_{t=1}^F \mathbb{E}_{\theta \sim p_{\text{data}}} [\log(\theta_t - 0)] + \sum_{t=1}^F \mathbb{E}_{\theta \sim p_{\mathbf{G}}} [\log(1 - \theta_t)]. \quad (3)$$

D_f constrains the recovered signals on a global spatial level, achieving the second term in (2).

2) **Aperture-Wise Attribute Discrimination:** When processing visual signals, the human visual system senses a scene based on salient attributes by associating captured attributes with memorized attributes [42]. Attributes in an image compose a visual distribution [43], and the distance between two distributions indicates their similarity in key attributes [32]. Attribute sensing is particularly crucial for ultrasound image recovery. Ultrasound images present inherent speckle patterns. They degrade image contrast and make recovery of anatomical attributes difficult [44], but serve as important natural tissue markers for motion tracking [45]. Therefore, the valuable speckle information is expected to be preserved as key attributes [46]. Different apertures induce different speckle patterns of the same imaging target. Besides, it is beneficial to consider contextual information of consecutive frames to preserve their temporal correlation. Therefore, in the second stage, D_a discriminates signals on the attribute level within a single sub-aperture across a time period.

The proposed discriminator regards D_f as an encoder. D_a is a decoder comprised of three symmetric convolutional layers, where the channel number of the last layer is K . It takes spatial feature maps of all frames from D_f as input and rearranges them from a frame-wise manner to an aperture-wise manner: $F \times \gamma_t \in \mathbb{R}^{N \times H_\gamma \times W_\gamma} \mapsto N \times \phi_n \in \mathbb{R}^{F \times H_\gamma \times W_\gamma}$, where ϕ_n denotes a rearranged feature map located in the n -th sub-aperture, $H_\gamma \times$

W_γ is the size of the feature maps. Specifically, F feature maps extracted by $\mathbf{D}_f$ are equally split along the aperture direction into N parts for N sub-apertures. All feature maps pertaining to one sub-aperture are stacked along the frame direction to be the input of $\mathbf{D}_a$.

Given features ϕ_n of a sub-aperture, $\mathbf{D}_a$ decodes K attribute maps $\mathcal{A}_n = [\mathcal{A}_n^k \mid k = 1 \sim K]$. Similarly, $\mathbf{D}_a$ predicts attribute maps for all sub-apertures and yields sigmoid-activated outputs: $[\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_N] \in \mathbb{R}^{N \times K \times H \times W}$. A mean-sum training strategy is proposed to implement adversarial learning based on the attribute distributions.

a) Mean: As formulated in (4), for the n -th sub-aperture, the discrete attribute distribution $\mathcal{V}_n$ is constructed by computing the mean value of each attribute map via global average pooling and concatenating the resultant K statistics. To create the real and fake labels for those attribute distributions, two K -element discrete Gaussian distributions with positive skew ($\mathcal{S}^+$) and negative skew ($\mathcal{S}^-$) are randomly sampled, respectively. Specifically, we set the shape parameter as ± 5 to generate skewed distributions and normalize them to be probability mass functions (Fig. 2(d)).

MSAF adopts KL divergence to measure the distance between $\mathcal{V}_n$ and the label distributions. Thus, the objective function of $\mathbf{D}_a$ can be formulated as

$$\begin{aligned} \mathcal{V}_n = \{ \overline{\mathcal{A}_n^1}, \overline{\mathcal{A}_n^2}, \dots, \overline{\mathcal{A}_n^K} \}, \\ \max_{\mathbf{G}} \min_{\mathbf{D}_a} \sum_{n=1}^N \mathbb{E}_{\mathcal{V} \sim p_{\mathbf{G}}} [\mathcal{D}_{\text{KL}}(\mathcal{S}^- \parallel \mathcal{V}_n)] \\ + \sum_{n=1}^N \mathbb{E}_{\mathcal{V} \sim p_{\text{data}}} [\mathcal{D}_{\text{KL}}(\mathcal{S}^+ \parallel \mathcal{V}_n)], \end{aligned} \quad (4)$$

where $\mathcal{D}_{\text{KL}}$ denotes the KL divergence. The generator aims to maximize the distance between the predicted distribution $\mathcal{V}_n$ coming from recovered signals and the fake label $\mathcal{S}^-$; In contrast, $\mathbf{D}_a$ aims to minimize that distance and also to minimize the distance between $\mathcal{V}_n$ coming from real signals and the real label $\mathcal{S}^+$.

b) Sum: Furthermore, a summing strategy ensures that the predicted attribute maps can represent local visual components within one sub-aperture along a time series. For each sub-aperture, $\mathbf{D}_a$ sums up K attribute maps and averages F consecutive beamformed frames in the corresponding sub-aperture. Then, it optimizes the summed attribute map to approximate the averaged image. Thus, an additional objective function can be formulated as

$$\min_{\mathbf{G}, \mathbf{D}_a} \sum_{n=1}^N \mathbb{E}_{\mathcal{A} \sim p_{\mathbf{G}}, p_{\text{data}}} \left\| \sum_{k=1}^K \mathcal{A}_n^k - \frac{1}{F} \sum_{t=1}^F \mathbf{I}_n^t \right\|_2^2, \quad (5)$$

where $\mathbf{I}_n^t$ denotes the t -th log-compressed envelope image generated from the n -th sub-aperture. Thus, the extracted attribute maps should be regarded as important local features summed over a time period as a sub-aperture image.

Using this mean-sum training strategy, MSAF recovers signals in each sub-aperture with local spatial details and temporal correlation, achieving the first term in (2).

C. Complex Spatiotemporal Generator

MSAF's generator consists of two modules: a complex UNet ($\mathbf{G}_u$) and a cascaded complex bidirectional ConvLSTM ($\mathbf{G}_l$). The former extracts the spatial features of each frame to recover the missing sub-aperture IQ signals. The latter extracts the temporal features across consecutive frames to refine the recovered IQ sequences further. Herein, the recovered IQ signals for missing sub-apertures are $\tilde{\mathbf{x}} = \mathbf{G}_l(\mathbf{G}_u(\mathbf{y}) + \mathbf{y})$.

1) Complex Neural Network: Our generator is entirely constructed by complex-valued convolutional neural networks dedicated to IQ data [38]. Let $\mathcal{X} = \mathcal{X}_{\text{real}} + j\mathcal{X}_{\text{imag}}$ be the input of a convolution operation. The output is calculated as

$$\begin{aligned} \begin{bmatrix} \mathcal{Y}_{\text{real}} \\ \mathcal{Y}_{\text{imag}} \end{bmatrix} &= \begin{bmatrix} \mathbf{W}_{\text{real}} & -\mathbf{W}_{\text{imag}} \\ \mathbf{W}_{\text{imag}} & \mathbf{W}_{\text{real}} \end{bmatrix} * \begin{bmatrix} \mathcal{X}_{\text{real}} \\ \mathcal{X}_{\text{imag}} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{W}_{\text{real}} * \mathcal{X}_{\text{real}} - \mathbf{W}_{\text{imag}} * \mathcal{X}_{\text{imag}} \\ \mathbf{W}_{\text{real}} * \mathcal{X}_{\text{imag}} + \mathbf{W}_{\text{imag}} * \mathcal{X}_{\text{real}} \end{bmatrix}, \end{aligned} \quad (6)$$

where $\mathbf{W}_{\text{real}}$ and $\mathbf{W}_{\text{imag}}$ are two sets of parameters for real (shaded boxes in Fig. 2(b)) and imaginary (dashed boxes in Fig. 2(b)) parts of data, respectively.

The complex UNet is built with seven 2D complex convolutional layers, including three encoder layers, a bottleneck layer, and three symmetrical decoder layers. These layers have a kernel size of 3×3 and channel numbers of 32, 64, 128, 256, 128, 64, and N , respectively. In the encoder, complex max pooling is applied after convolution to downsample the complex feature maps by $\frac{1}{2}$. Different from standard UNet [47], $\mathbf{G}_u$ applies the Leaky ReLU activation function, ensuring gradient propagation in negative IQ values. The input of $\mathbf{G}_u$ takes sparse IQ signals, denoted as $\mathcal{X} \in \mathbb{C}^{F \times N \times H \times W}$. $\mathbf{G}_u$ goes through F frames as a batch and recovers missing sub-apertures based on 2D spatial features.

2) Complex Bidirectional ConvLSTM: To leverage ultrafast ultrasound for highly dynamic tissues, we propose a complex bidirectional ConvLSTM to capture temporal features along the slow time (i.e., frame) direction by long short-term memory. $\mathbf{G}_l$ is a gated recurrent neural network and considers F frames as time steps $\{\mathcal{X}_t \mid t = 1 \sim F\}$.

$\mathbf{G}_l$ is constituted of three layers, where F units make up a layer. A unit contains a memory cell $\mathcal{C}_t$, an input gate i_t , a forget gate f_t , an output gate o_t , and an output state $\mathcal{H}_t$. Each unit can be formulated as follows:

$$\begin{aligned} i_t &= \sigma(\mathbf{W}_{xi} * \mathcal{X}_t + \mathbf{W}_{hi} * \mathcal{H}_{t-1} + \mathbf{W}_{ci} * \mathcal{C}_{t-1} + b_i) \\ f_t &= \sigma(\mathbf{W}_{xf} * \mathcal{X}_t + \mathbf{W}_{hf} * \mathcal{H}_{t-1} + \mathbf{W}_{cf} * \mathcal{C}_{t-1} + b_f) \\ \mathcal{C}_t &= f_t \circ \mathcal{C}_{t-1} + i_t \circ \tau(\mathbf{W}_{xc} * \mathcal{X}_t + \mathbf{W}_{hc} * \mathcal{H}_{t-1} + b_c) \\ o_t &= \sigma(\mathbf{W}_{xo} * \mathcal{X}_t + \mathbf{W}_{ho} * \mathcal{H}_{t-1} + \mathbf{W}_{co} * \mathcal{C}_t + b_o) \\ \mathcal{H}_t &= o_t \circ \tau(\mathcal{C}_t), \end{aligned} \quad (7)$$

where σ and τ denote complex sigmoid function and complex tanh function, respectively; $*$ and $\circ$ denote complex convolution as (6) and complex Hadamard product, respectively. For the subscripts of complex convolutional weights $\mathbf{W}$ and bias b ,

TABLE I
THE IMAGING PARAMETERS FOR IN VIVO DATA ACQUISITIONS

Imaging Parameter	Value
Probe	P4-2
Wave mode	Diverging wave with $\frac{\pi}{2}$ opening angle
Center/transmit frequency	2.5 MHz
No. sub-apertures	8
Sub-aperture width	2.56 mm
No. elements	64
Element pitch	0.32 mm
Virtual source location	1.28 mm behind the array
Excitation	CaSA [33] driven at 4V
Acquisition Framerate	1600 fps

the first index denotes the current gate, while the second index denotes the target gate. Each unit updates the current state C_t and hidden state $\mathcal{H}_t$ via (7) based on memory from the $(t-1)$ -th step and information from the t -th step. Thus, it captures the stable temporal feature between frames and is resistant to random noise.

Bidirectional $\mathbf{G}_l$ features both $\overrightarrow{\text{forward}}$ and $\overleftarrow{\text{backward}}$ paths to process the frame sequence in opposite directions and then decides the output by data dependencies between two directions. The final output $\tilde{\mathbf{x}} \in \mathbb{C}^{F \times N \times H \times W}$ can be calculated as

$$\tilde{\mathbf{x}} = \mathbf{F}_{cat} \left[\tau \left(\overrightarrow{\mathbf{W}}_t * \overrightarrow{\mathcal{H}}_t + \overleftarrow{\mathbf{W}}_t * \overleftarrow{\mathcal{H}}_t + b_t \right) \right], \quad (8)$$

where $\overrightarrow{\mathcal{H}}_t$ and $\overleftarrow{\mathcal{H}}_t$ are final hidden states in forward and backward directions, respectively; $\overrightarrow{\mathbf{W}}_t$ and $\overleftarrow{\mathbf{W}}_t$ are complex convolutional weights; b_t is complex bias; $\mathbf{F}_{cat}$ denotes concatenation of F recovered frames along the slow time axis. $\mathbf{G}_l$ captures temporal features and learns inter-frame correlation, further recovering IQ signals that encode tissue dynamics.

D. Domain Converter

IQ signals can be represented in Cartesian (real and imaginary parts) or polar (amplitude and phase) form. The generator processes IQ signals in native Cartesian representation using a complex-valued neural network. In ultrasound imaging, the signal amplitude and phase are used for B-mode display and motion estimation, respectively [48]. To align adversarial discrimination with physical interpretation of ultrasound signals, a domain converter (Fig. 2(e)) is proposed to transform $\mathbf{G}_l$'s Cartesian outputs into polar form before evaluation by $\mathbf{D}_f$, bridging the generator and discriminator.

IV. EXPERIMENTS

A. Data Acquisition and Processing

This study was conducted on a large synthetic aperture ultrasound dataset of in vivo human hearts. The cohort included 74 healthy subjects and 21 subjects with hypertension and/or diabetes. Raw IQ signals were acquired using a Verasonics Vantage 256 system (Verasonics, Kirkland, WA, USA) with a P4-2 phased array probe operating at 2.5 MHz [49]. The array probe was divided into eight sub-apertures, each comprising eight elements. Table I lists imaging parameters. Six standard views

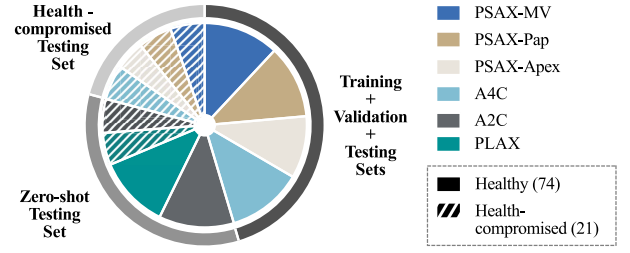


Fig. 3. Distribution of cardiac ultrasound data in six imaging views acquired from 74 healthy and 21 health-compromised human subjects: healthy datasets (training:validation:testing = 7:2:1), a health-compromised testing set, and a zero-shot testing set.

of echocardiography were acquired for each subject, including the apical four-chamber (A4C) and two-chamber (A2C) views, the parasternal short-axis views at the mitral valve (PSAX-MV), papillary muscle (PSAX-Pap), and apical (PSAX-Apex) levels, as well as the parasternal long-axis view (PLAX) [50]. Fig. 3 shows the data distribution, where the numbers of frames for the six echocardiographic views were 16.4 k, 16 k, 13.6 k, 16.48 k, 16.24 k, and 15.59 k, respectively.

The dataset was partitioned into three parts (Fig. 3). To validate the generalizability on non-healthy samples, we constructed an independent test set comprised of only chronic condition data, ensuring no training exposure to non-healthy cases. The A2C and PLAX views were also excluded from training and served as two out-of-view testing sets to assess zero-shot generalizability. Data from the remaining echocardiographic views of the healthy subjects were randomly sampled into training, validation, and testing sets in a ratio of 7:2:1, on a subject-by-subject basis. After training on the cardiac dataset, we applied MSAF to an in vivo abdominal ultrasound dataset, including 4800 frames of normal human liver and kidney.

The received raw signals were first transformed to in-phase and quadrature data, with dimensions of $H \times W = 320 \times 64$ per 2D frame per sub-aperture. Then, the full-aperture beam-formed IQ signals were split into odd-aperture signals as input $\mathbf{y} \in \mathbb{C}^{F \times 4 \times 320 \times 64}$ and even-aperture signals as ground truth $\tilde{\mathbf{x}} \in \mathbb{C}^{F \times 4 \times 320 \times 64}$. In MSAF, we configured $F = 5$, which captured temporal features across five frames. To investigate the effect of temporal window lengths on temporal sensing of fast cardiac dynamics, we tested four temporal windows with each containing five uniformly sampled frames: 25 ms, 50 ms, 75 ms, and 100 ms.

B. Benchmark Models

To the best of our knowledge, this is the first deep-learning research for the ultrasound sparse-aperture signal recovery. Although there are no existing comparison models, we conducted a benchmarking experiment by modifying six CNN-based and three transformer-based models to fit our task. These benchmark models are categorized into seven real-valued and two complex-valued models. Since real-valued models lack inherent mechanisms to extract the interrelationship between in-phase and quadrature components in the complex domain, all inputs for

TABLE II
DATA FORMAT AND COMPUTATIONAL LOAD OF MSAF AND NINE BENCHMARK MODELS

Model	Data Format			Computational Load			
	Domain	Frame	Dimension	#Params	FLOPs	Inference Time	GPU Memory
UNet [24]	RF	1	$\mathbb{R}^{1 \times 4 \times 1280 \times 64}$	8.6 M	142.19 G	27.24 ms	3364 MB
GAN [34]	RF	1	$\mathbb{R}^{1 \times 4 \times 1280 \times 64}$	11.4 M [†]	190.63 G [†]	27.24 ms	3364 MB
TransUNet [35]	RF	1	$\mathbb{R}^{1 \times 4 \times 1280 \times 64}$	41.4 M	129.03 G	43.95 ms	2002 MB
SwinUNet [36]	RF	1	$\mathbb{R}^{1 \times 4 \times 1280 \times 64}$	62.0 M	56.96 G	35.05 ms	1349 MB
Restormer [37]	RF	1	$\mathbb{R}^{1 \times 4 \times 1280 \times 64}$	25.3 M	128.01 G	143.05 ms	2406 MB
Complex UNet [38]	IQ	1	$\mathbb{C}^{1 \times 4 \times 320 \times 64}$	17.3 M	48.22 G	41.99 ms	908 MB
Dual-branch UNet [39]	IQ	1	$\mathbb{R}^{2 \times 4 \times 320 \times 64}$	17.3 M	55.11 G	12.71 ms	1022 MB
3D UNet [40]	RF	5	$\mathbb{R}^{5 \times 4 \times 1280 \times 64}$	19.1 M	44.53 G	78.18 ms	11586 MB
Bi-ConvLSTM [41]	RF	5	$\mathbb{R}^{5 \times 4 \times 1280 \times 64}$	5.0 M	254.95 G	123.59 ms	2570 MB
MSAF	IQ	5	$\mathbb{C}^{5 \times 4 \times 320 \times 64}$	25.6 M [‡]	430.64 G [‡]	174.43 ms	1654 MB

[†] included 2.8 M parameters and 48.44 G FLOPs for the discriminator.

[‡] included 7.2 M parameters and 127.47 G FLOPs for the discriminator.

these models were converted from IQ to radio-frequency (RF) signal format. Note that the RF signals had an axial dimension 4 times larger than the IQ signals because of their higher sampling rate requirements.

Real-valued models: (1) **UNet** [24], the most prominent CNN architecture for medical image processing; (2) **GAN** [34], the most prevalent adversarial learning architecture; (3) **TransUNet** [35], a leading hybrid CNN-Transformer network; (4) **SwinUNet** [36], a widely recognized pure-Transformer for medical images; (5) **Restormer** [37], an efficient transformer for high-resolution restoration. The above five models process one RF frame per inference. (6) **3D UNet** [40], a UNet with 3D convolution to process five stacked RF frames; (7) **Bi-ConvLSTM** [41], a bi-directional convolutional long short-term memory network to process five sequential RF frames.

Complex-valued models: (8) **Complex UNet** [38], a complex-valued neural network to process a single IQ frame per inference using complex convolution; (9) **Dual-branch UNet** [39], a UNet handling real and imaginary parts separately through two parallel branches. Table II summarizes the signal domain, the number of input frames, and input dimension for each benchmark model. The activation function in all models was set to be Leaky ReLU to process negative values. All models were retrained on our dataset with the same hyperparameters as MSAF.

C. Implementation Details

All scripts were written with Python 3.8, Pytorch 1.9.¹ Experiments were conducted on two 24 G NVIDIA RTX 3090 GPUs and an Intel Xeon CPU. The Adam optimizer configured with a batch size of 8 and a learning rate of 1×10^{-4} was used for training. The attribute number K for aperture-wise attribute discrimination was set to 10. Peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), mean absolute percentage error (MAPE), contrast-to-noise ratio (CNR), and contrast ratio (CR) were used to evaluate recovery performance.

¹ The codes will be released on <https://github.com/JustinJZhang/MSAF/tree/main>.

V. RESULTS

A. Comparison Study

Table III summarizes the recovery performances in terms of PSNR and SSIM on four echocardiographic views. Both metrics were calculated on the recovered IQ (real and imaginary parts) or RF signals. Our MSAF achieved the best PSNR values of 34.21, 34.18, 34.49, and 33.47 dB on the PSAX-MV, PSAX-Pap, PSAX-Apex, and A4C views, respectively. Averaged across views, MSAF exhibited 18.64%, 19.71%, 9.24%, 16.88%, 9.39%, 12.31%, 9.64%, 12.34%, and 11.36% PSNR improvements over UNet, 3D UNet, Bi-ConvLSTM, GAN, TransUNet, SwinUNet, Restormer, Dual-branch UNet, and Complex UNet, respectively. Regarding structural fidelity, MSAF achieved SSIM values of 0.92, 0.92, 0.93, and 0.86 on PSAX-MV, PSAX-Pap, PSAX-Apex, and A4C views, respectively. View-averaged SSIMs from MSAF were 12.37%, 13.09%, 4.04%, 10.45%, 5.26%, 7.81%, 4.62%, 5.67%, and 5.23% higher than those from the above models, respectively. Evaluations in the signal domain validated MSAF's superiority in terms of high-quality raw signal recovery.

Moreover, we assessed the B-mode images produced from the recovered IQ or RF signals. Table III shows that MSAF achieved the best PSNRs of 31.82, 31.76, 31.66, 30.83 dB, and the best SSIMs of 0.88, 0.88, 0.88, 0.78 in the PSAX-MV, PSAX-Pap, PSAX-Apex, and A4C views, respectively. Fig. 4 shows B-mode images of one recovered sub-aperture from MSAF and nine benchmark models in the above four views. It demonstrated that MSAF produced B-mode images closest to the ground truths (GTs), preserving the speckle pattern and rich details. In contrast, other models only recovered dominant structures while failing to capture subtle image attributes. This limitation is exemplified in Fig. 4(a) when compared with Fig. 8 (Attribute #3). Other models sensed and recovered solely the strongest signal component, overlooking weaker yet physiologically significant attributes that constituted full tissue characteristics. Therefore, MSAF is able to recover finer B-mode images with more details compared to the other models.

Fig. 5 shows the error distributions resulting from different models. Averaged from IQ and B-mode domains, MSAF produced MAPEs of 21%, 14%, 17%, and 17% in the A4C,

TABLE III
THE PSNR AND SSIM VALUES ON THE RECOVERED IQ (REAL AND IMAGINARY PARTS) OR RF SIGNALS, AND B-MODE IMAGES FROM MSAF AND NINE BENCHMARK MODELS

View	Domain	Model	PSNR (dB)		SSIM			
			Recovered IQ/RF Signal		B-Mode	Recovered IQ/RF Signal		B-Mode
PSAX-MV	RF	UNet [24]	29.32 $\pm$ 0.25		27.28 $\pm$ 0.26	0.82 $\pm$ 0.01		0.58 $\pm$ 0.01
		3D UNet [40]	28.76 $\pm$ 0.24		24.06 $\pm$ 0.25	0.81 $\pm$ 0.01		0.55 $\pm$ 0.03
		Bi-ConvLSTM [41]	31.83 $\pm$ 0.38		27.74 $\pm$ 0.32	0.88 $\pm$ 0.01		0.83 $\pm$ 0.03
		GAN [34]	29.46 $\pm$ 0.25		24.70 $\pm$ 0.25	0.83 $\pm$ 0.01		0.55 $\pm$ 0.03
		TransUNet [35]	31.61 $\pm$ 0.49		27.48 $\pm$ 0.55	0.87 $\pm$ 0.01		0.67 $\pm$ 0.03
		SwinUNet [36]	30.31 $\pm$ 0.35		26.41 $\pm$ 0.37	0.84 $\pm$ 0.01		0.66 $\pm$ 0.03
		Restormer [37]	31.57 $\pm$ 0.36		27.51 $\pm$ 0.46	0.88 $\pm$ 0.01		0.68 $\pm$ 0.01
	IQ	Dual-branch UNet [39]	30.64 $\pm$ 0.44 [†]	30.69 $\pm$ 0.41 [‡]	27.50 $\pm$ 0.28	0.87 $\pm$ 0.01 [†]	0.87 $\pm$ 0.01 [‡]	0.66 $\pm$ 0.02
		Complex UNet [38]	30.93 $\pm$ 0.39 [†]	30.97 $\pm$ 0.36 [‡]	27.34 $\pm$ 0.30	0.87 $\pm$ 0.01 [†]	0.87 $\pm$ 0.01 [‡]	0.67 $\pm$ 0.02
		MSAF	33.91 $\pm$ 0.35[†]	34.51 $\pm$ 0.39[‡]	31.82 $\pm$ 0.21	0.92 $\pm$ 0.01[†]	0.92 $\pm$ 0.01[‡]	0.88 $\pm$ 0.01
PSAX-Pap	RF	UNet [24]	29.16 $\pm$ 0.25		27.06 $\pm$ 0.25	0.81 $\pm$ 0.02		0.57 $\pm$ 0.01
		3D UNet [40]	28.64 $\pm$ 0.25		24.15 $\pm$ 0.25	0.80 $\pm$ 0.02		0.55 $\pm$ 0.05
		Bi-ConvLSTM [41]	31.37 $\pm$ 0.39		27.63 $\pm$ 0.30	0.87 $\pm$ 0.02		0.82 $\pm$ 0.05
		GAN [34]	29.40 $\pm$ 0.26		24.52 $\pm$ 0.27	0.82 $\pm$ 0.02		0.55 $\pm$ 0.05
		TransUNet [35]	31.11 $\pm$ 0.45		26.77 $\pm$ 0.51	0.86 $\pm$ 0.02		0.62 $\pm$ 0.05
		SwinUNet [36]	30.32 $\pm$ 0.38		26.12 $\pm$ 0.41	0.84 $\pm$ 0.02		0.65 $\pm$ 0.05
		Restormer [37]	31.43 $\pm$ 0.41		27.10 $\pm$ 0.58	0.88 $\pm$ 0.02		0.69 $\pm$ 0.01
	IQ	Dual-branch UNet [39]	30.56 $\pm$ 0.49 [†]	30.58 $\pm$ 0.45 [‡]	27.41 $\pm$ 0.30	0.86 $\pm$ 0.02 [†]	0.86 $\pm$ 0.02 [‡]	0.66 $\pm$ 0.04
		Complex UNet [38]	30.84 $\pm$ 0.44 [†]	30.86 $\pm$ 0.41 [‡]	27.24 $\pm$ 0.33	0.87 $\pm$ 0.02 [†]	0.87 $\pm$ 0.02 [‡]	0.66 $\pm$ 0.04
		MSAF	33.89 $\pm$ 0.40[†]	34.46 $\pm$ 0.45[‡]	31.76 $\pm$ 0.22	0.92 $\pm$ 0.01[†]	0.92 $\pm$ 0.01[‡]	0.88 $\pm$ 0.01
PSAX-Apex	RF	UNet [24]	28.70 $\pm$ 0.29		24.16 $\pm$ 0.12	0.81 $\pm$ 0.01		0.55 $\pm$ 0.04
		3D UNet [40]	28.90 $\pm$ 0.11		24.27 $\pm$ 0.07	0.81 $\pm$ 0.01		0.57 $\pm$ 0.01
		Bi-ConvLSTM [41]	31.55 $\pm$ 0.08		27.51 $\pm$ 0.08	0.88 $\pm$ 0.01		0.85 $\pm$ 0.01
		GAN [34]	29.59 $\pm$ 0.10		24.79 $\pm$ 0.08	0.83 $\pm$ 0.01		0.59 $\pm$ 0.01
		TransUNet [35]	32.58 $\pm$ 0.34		28.22 $\pm$ 0.24	0.89 $\pm$ 0.01		0.69 $\pm$ 0.01
		SwinUNet [36]	31.85 $\pm$ 0.31		28.21 $\pm$ 0.24	0.87 $\pm$ 0.01		0.74 $\pm$ 0.01
		Restormer [37]	31.44 $\pm$ 0.13		28.09 $\pm$ 0.16	0.88 $\pm$ 0.01		0.87 $\pm$ 0.01
	IQ	Dual-branch UNet [39]	30.85 $\pm$ 0.09 [†]	30.80 $\pm$ 0.09 [‡]	27.37 $\pm$ 0.08	0.87 $\pm$ 0.01 [†]	0.87 $\pm$ 0.01 [‡]	0.69 $\pm$ 0.01
		Complex UNet [38]	31.08 $\pm$ 0.08 [†]	31.07 $\pm$ 0.09 [‡]	27.20 $\pm$ 0.10	0.88 $\pm$ 0.01 [†]	0.88 $\pm$ 0.01 [‡]	0.71 $\pm$ 0.01
		MSAF	34.14 $\pm$ 0.12[†]	34.83 $\pm$ 0.13[‡]	31.66 $\pm$ 0.05	0.92 $\pm$ 0.01[†]	0.93 $\pm$ 0.01[‡]	0.88 $\pm$ 0.01
A4C	RF	UNet [24]	27.74 $\pm$ 0.30		25.67 $\pm$ 0.38	0.78 $\pm$ 0.02		0.38 $\pm$ 0.03
		3D UNet [40]	27.59 $\pm$ 0.32		22.79 $\pm$ 0.33	0.77 $\pm$ 0.02		0.36 $\pm$ 0.01
		Bi-ConvLSTM [41]	30.06 $\pm$ 0.35		26.38 $\pm$ 0.31	0.83 $\pm$ 0.02		0.81 $\pm$ 0.03
		GAN [34]	28.19 $\pm$ 0.32		23.39 $\pm$ 0.34	0.79 $\pm$ 0.02		0.37 $\pm$ 0.03
		TransUNet [35]	29.33 $\pm$ 0.40		25.25 $\pm$ 0.44	0.82 $\pm$ 0.02		0.56 $\pm$ 0.04
		SwinUNet [36]	28.92 $\pm$ 0.37		24.74 $\pm$ 0.38	0.81 $\pm$ 0.02		0.54 $\pm$ 0.03
		Restormer [37]	29.91 $\pm$ 0.39		25.60 $\pm$ 0.45	0.83 $\pm$ 0.02		0.55 $\pm$ 0.01
	IQ	Dual-branch UNet [39]	29.29 $\pm$ 0.37 [†]	29.32 $\pm$ 0.36 [‡]	26.05 $\pm$ 0.34	0.83 $\pm$ 0.02 [†]	0.83 $\pm$ 0.02 [‡]	0.58 $\pm$ 0.01
		Complex UNet [38]	29.52 $\pm$ 0.36 [†]	29.57 $\pm$ 0.36 [‡]	25.80 $\pm$ 0.35	0.83 $\pm$ 0.02 [†]	0.83 $\pm$ 0.02 [‡]	0.59 $\pm$ 0.01
		MSAF	33.20 $\pm$ 0.27[†]	33.73 $\pm$ 0.31[‡]	30.83 $\pm$ 0.29	0.85 $\pm$ 0.01[†]	0.86 $\pm$ 0.01[‡]	0.78 $\pm$ 0.02

[†] PSNR and SSIM of the real parts in IQ signals, [‡] PSNR and SSIM of the imaginary parts in IQ signals.

PSAX-MV, PSAX-Pap, and PSAX-Apex views, respectively, which were consistently the lowest among all models. Fig. 6 shows the CNR and CR results. In each view, the myocardium (heart walls) was chosen as the signal region, while the atrial or ventricular cavity at the same depth was selected as the background. MSAF yielded CNRs of 2.77, 4.28, 2.68, and 1.94dB in the above four views, where the CRs were 13.43, 19.35, 14.75, and 14.31dB, respectively. Compared with other models, MSAF consistently achieved the highest CNR and CR with a narrow interquartile range. The error and contrast analyses evidence MSAF's high-fidelity and high-contrast advantages over all benchmark models.

B. Ablation Study

MSAF is a multi-level sensing framework, comprising spatial-level sensing through a complex UNet (G_u), temporal-level sensing via a complex Bi-ConvLSTM (G_l), global-level sensing powered by frame-wise global discrimination

(D_f), and attribute-level sensing with aperture-wise attribute discrimination (D_a). In addition, a domain converter between the generator and the discriminator is proposed to enrich feature representation. To verify the effectiveness of these modules, five ablation models were built by eliminating each of them from MSAF.

Table IV summarizes PSNR and SSIM obtained by the five ablation models. In the IQ domain, the view-averaged PSNR of the complete MSAF was 7%, 8%, 12%, 8%, and 6% higher than the MSAF without D_f , MSAF without D_a , MSAF without G_u , MSAF without G_l , and MSAF without domain converter, respectively. The averaged SSIM of the complete MSAF was 4%, 5%, 15%, 4%, and 3% higher than the aforementioned ablation models.

For B-mode images, the average SSIM of the complete MSAF was 6%, 8%, 10%, 9%, and 7% higher than the above five ablation models. The average SSIM of the complete MSAF was 3%, 3%, 8%, 4%, and 2% higher than the five ablation models. These results indicated that all four proposed sensing

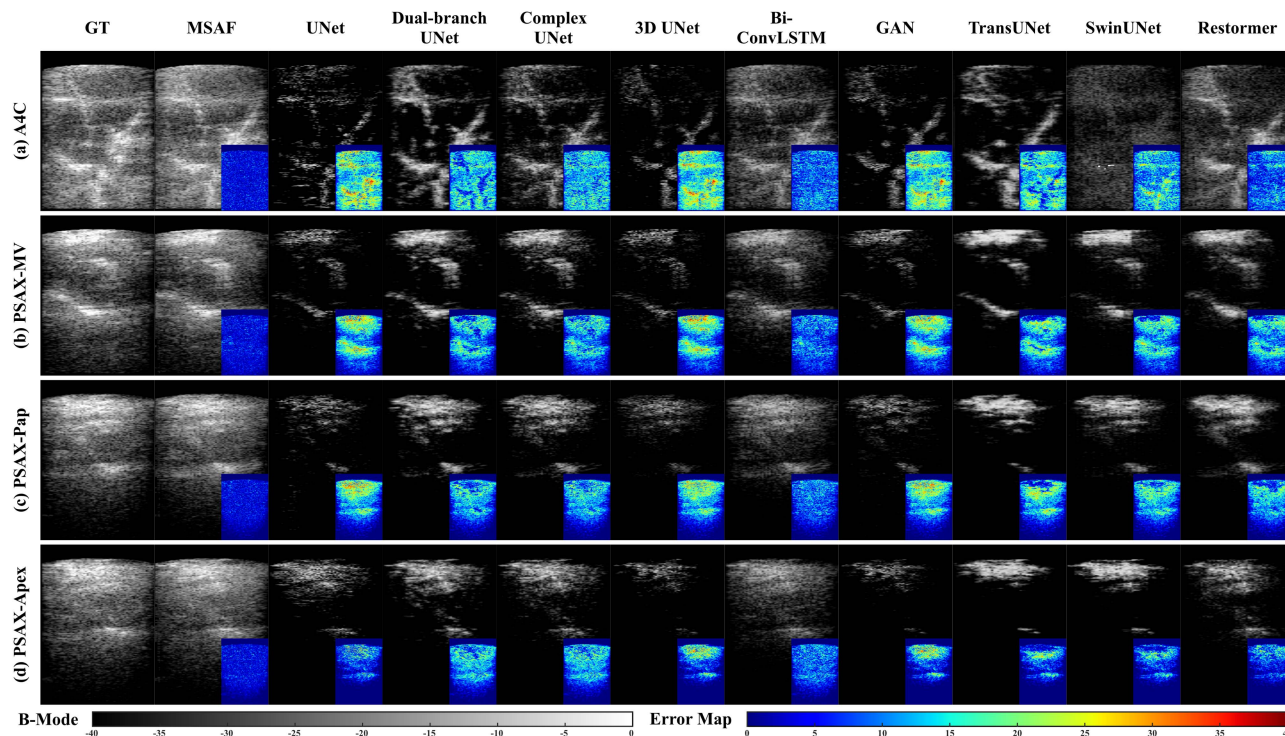


Fig. 4. B-mode images of one sub-aperture recovered by various deep learning models, where the recovered signals were log-compressed and displayed in a dynamic range of 40 dB. (a)–(d) show in vivo ultrasound B-mode images of the heart in the A4C, PSAX-MV, PSAX-Pap, and PSAX-Apex views. The first column shows the ground truth, which was used for obtaining error maps (color images on the bottom right corner); The remaining columns show B-mode images generated using the signals recovered by MSAF and nine benchmark models.

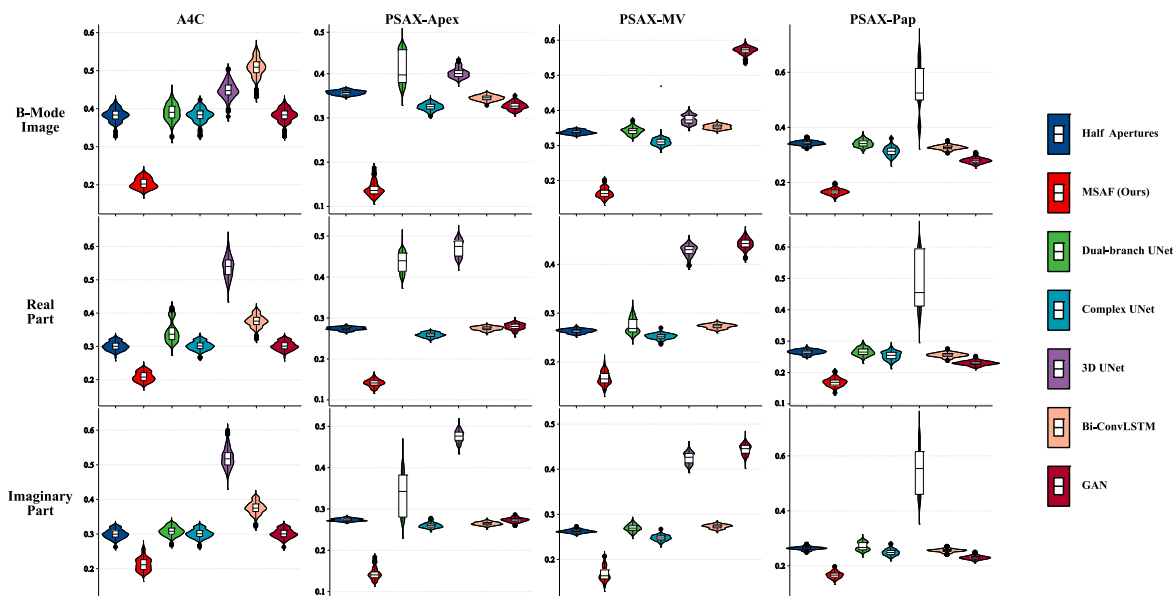


Fig. 5. The violin plots of mean absolute percentage errors (MAPE) with respect to full-aperture signals. Note that UNet errors are excluded because of their significantly higher values than the others.

levels contributed distinctively to the IQ signal recovery in synthetic aperture ultrasound, and the current framework achieved superior recovery performance.

C. Temporal Window Analysis

Table V summarizes the results from four different temporal windows and demonstrates improvement in recovery quality

TABLE IV

THE PSNR AND SSIM VALUES ON THE RECOVERED IQ (REAL AND IMAGINARY PARTS) SIGNALS AND B-MODE IMAGES FROM MSAF AND FIVE ABLATION MODELS

View	Ablation Model	PSNR (dB)			SSIM		
		Real	Imaginary	B-Mode	Real	Imaginary	B-Mode
PSAX-MV	MSAF	33.91 $\pm$ 0.35	34.51 $\pm$ 0.39	31.82 $\pm$ 0.21	0.92 $\pm$ 0.01	0.92 $\pm$ 0.01	0.88 $\pm$ 0.01
	w/o $\mathbf{D}_f$	30.99 $\pm$ 0.17	31.52 $\pm$ 0.17	28.50 $\pm$ 0.10	0.89 $\pm$ 0.01	0.88 $\pm$ 0.01	0.84 $\pm$ 0.01
	w/o $\mathbf{D}_a$	30.93 $\pm$ 0.28	31.41 $\pm$ 0.27	29.53 $\pm$ 0.10	0.87 $\pm$ 0.02	0.88 $\pm$ 0.01	0.85 $\pm$ 0.01
	w/o $\mathbf{G}_u$	28.84 $\pm$ 0.09	29.58 $\pm$ 0.12	27.48 $\pm$ 0.07	0.62 $\pm$ 0.17	0.79 $\pm$ 0.02	0.78 $\pm$ 0.01
	w/o $\mathbf{G}_l$	30.94 $\pm$ 0.19	31.49 $\pm$ 0.19	28.63 $\pm$ 0.12	0.88 $\pm$ 0.01	0.87 $\pm$ 0.01	0.84 $\pm$ 0.01
	w/o Domain Converter	31.11 $\pm$ 0.16	31.60 $\pm$ 0.15	29.71 $\pm$ 0.12	0.83 $\pm$ 0.11	0.88 $\pm$ 0.01	0.85 $\pm$ 0.01
PSAX-Pap	MSAF	33.89 $\pm$ 0.40	34.46 $\pm$ 0.45	31.76 $\pm$ 0.22	0.92 $\pm$ 0.01	0.92 $\pm$ 0.01	0.88 $\pm$ 0.01
	w/o $\mathbf{D}_f$	31.69 $\pm$ 0.23	32.17 $\pm$ 0.22	30.25 $\pm$ 0.17	0.87 $\pm$ 0.09	0.88 $\pm$ 0.01	0.86 $\pm$ 0.01
	w/o $\mathbf{D}_a$	31.52 $\pm$ 0.56	32.04 $\pm$ 0.56	29.94 $\pm$ 0.30	0.89 $\pm$ 0.02	0.88 $\pm$ 0.02	0.86 $\pm$ 0.01
	w/o $\mathbf{G}_u$	30.02 $\pm$ 0.08	30.28 $\pm$ 0.07	28.96 $\pm$ 0.09	0.82 $\pm$ 0.08	0.85 $\pm$ 0.01	0.83 $\pm$ 0.01
	w/o $\mathbf{G}_l$	31.57 $\pm$ 0.14	32.14 $\pm$ 0.13	28.86 $\pm$ 0.09	0.91 $\pm$ 0.01	0.88 $\pm$ 0.01	0.85 $\pm$ 0.01
	w/o Domain Converter	32.65 $\pm$ 0.29	33.20 $\pm$ 0.30	29.51 $\pm$ 0.26	0.93 $\pm$ 0.01	0.90 $\pm$ 0.01	0.86 $\pm$ 0.01
PSAX-Apex	MSAF	34.14 $\pm$ 0.12	34.83 $\pm$ 0.13	31.66 $\pm$ 0.05	0.92 $\pm$ 0.01	0.93 $\pm$ 0.01	0.88 $\pm$ 0.01
	w/o $\mathbf{D}_f$	30.85 $\pm$ 0.15	31.58 $\pm$ 0.14	29.11 $\pm$ 0.14	0.89 $\pm$ 0.02	0.88 $\pm$ 0.01	0.85 $\pm$ 0.01
	w/o $\mathbf{D}_a$	30.37 $\pm$ 0.06	30.99 $\pm$ 0.06	27.78 $\pm$ 0.05	0.90 $\pm$ 0.01	0.87 $\pm$ 0.01	0.84 $\pm$ 0.01
	w/o $\mathbf{G}_u$	28.45 $\pm$ 0.01	28.92 $\pm$ 0.01	26.94 $\pm$ 0.03	0.72 $\pm$ 0.12	0.79 $\pm$ 0.02	0.79 $\pm$ 0.01
	w/o $\mathbf{G}_l$	30.38 $\pm$ 0.08	31.06 $\pm$ 0.08	27.67 $\pm$ 0.06	0.90 $\pm$ 0.01	0.87 $\pm$ 0.01	0.84 $\pm$ 0.01
	w/o Domain Converter	31.62 $\pm$ 0.19	32.29 $\pm$ 0.19	29.27 $\pm$ 0.16	0.90 $\pm$ 0.20	0.90 $\pm$ 0.01	0.86 $\pm$ 0.01
A4C	MSAF	33.20 $\pm$ 0.27	33.73 $\pm$ 0.31	30.83 $\pm$ 0.29	0.85 $\pm$ 0.01	0.86 $\pm$ 0.01	0.78 $\pm$ 0.02
	w/o $\mathbf{D}_f$	32.67 $\pm$ 0.31	33.23 $\pm$ 0.32	30.97 $\pm$ 0.32	0.83 $\pm$ 0.03	0.84 $\pm$ 0.02	0.77 $\pm$ 0.01
	w/o $\mathbf{D}_a$	31.59 $\pm$ 0.29	32.06 $\pm$ 0.30	29.07 $\pm$ 0.34	0.77 $\pm$ 0.07	0.79 $\pm$ 0.02	0.75 $\pm$ 0.01
	w/o $\mathbf{G}_u$	31.62 $\pm$ 0.49	32.25 $\pm$ 0.48	30.05 $\pm$ 0.40	0.80 $\pm$ 0.02	0.79 $\pm$ 0.02	0.76 $\pm$ 0.01
	w/o $\mathbf{G}_l$	31.72 $\pm$ 0.14	32.29 $\pm$ 0.15	29.08 $\pm$ 0.11	0.83 $\pm$ 0.01	0.79 $\pm$ 0.01	0.75 $\pm$ 0.01
	w/o Domain Converter	33.00 $\pm$ 0.14	33.53 $\pm$ 0.18	29.31 $\pm$ 0.16	0.84 $\pm$ 0.03	0.84 $\pm$ 0.01	0.77 $\pm$ 0.01

TABLE V

THE PSNR AND SSIM VALUES FOR THE RECOVERED IQ SIGNALS AND B-MODE IMAGES USING FOUR DIFFERENT TEMPORAL WINDOWS IN MSAF

Temporal Window		Domain	PSNR (dB)				SSIM			
			PSAX-MV	PSAX-Pap	PSAX-Apex	A4C	PSAX-MV	PSAX-Pap	PSAX-Apex	A4C
25 ms	5 Frames	IQ Signal	34.21	34.18	34.49	33.47	0.92	0.92	0.93	0.86
		B-Mode	31.82	31.76	31.66	30.83	0.88	0.88	0.88	0.78
50 ms	10 Frames	IQ Signal	34.63	34.54	34.56	34.73	0.92	0.92	0.92	0.93
		B-Mode	31.93	31.71	31.74	31.94	0.91	0.90	0.90	0.91
75 ms	15 Frames	IQ Signal	35.18	35.08	35.17	35.30	0.93	0.93	0.93	0.93
		B-Mode	32.03	31.75	31.80	32.04	0.91	0.90	0.90	0.91
100 ms	20 Frames	IQ Signal	35.27	35.13	35.27	35.38	0.94	0.93	0.94	0.94
		B-Mode	32.18	31.82	31.90	32.17	0.91	0.90	0.90	0.90

with longer temporal contexts. Expanding the temporal window from 25 ms to 50 ms, 75 ms, and 100 ms led to average PSNR gains of 1.56%, 3.22%, and 3.45% for IQ signals and 1.00%, 1.23%, and 1.58% for B-mode images, respectively.

Furthermore, we analyzed the learned temporal features using cosine similarity at both feature and frame levels. We examined a 100-ms cardiac event spanning a QRS complex in the ECG signal, which was cropped into the aforementioned four temporal windows (Fig. 7(a)). The heatmap in Fig. 7(a) shows the cosine similarity between the original frames, marking ventricular depolarization. In Fig. 7(b) and (c), the bottom row shows the cosine similarity between five time steps in $\mathbf{G}_l$. They shared similar patterns with Fig. 7(a) in the corresponding windows, confirming $\mathbf{G}_l$'s ability to capture cardiac dynamics in different scales; the top row shows similarities among the learned feature space, where the average cosine similarity followed 75 ms > 100 ms > 50 ms > 25 ms. This trend aligned with the time-step similarity ranking, where lower temporal correlation reflected higher dynamics and thus requested sophisticated features with lower similarity.

D. Aperture-Wise Attribute Distribution

Fig. 8 exemplifies the attribute maps of $\mathbf{D}_a$ in the A4C view. For each sub-aperture, $\mathbf{D}_a$ decoded $K = 10$ attribute maps from the input. According to MSAF's mean-sum strategy, red curves showed the randomly generated distribution labels. Histograms showed the distribution formed by attribute maps, where the value of one bin was equal to the average of an attribute map. Fig. 8 shows that the outputted real and fake distributions were close to the real and fake labels, respectively. Furthermore, 10 attributes can be summed as complete signals regardless of real or fake signals. Different attributes can therefore represent different local components of the complete signals. A higher bin denoted an attribute with stronger echo signals, while a lower bin denoted an attribute with weaker signals. The difference between fake and real attributes showed how $\mathbf{D}_a$ discriminated the signals given a learned attribute. For example, the 3-rd and 8-th attributes respectively had the highest bins in the real and fake distributions. They represented the strongest echo signals from the atrial and ventricular walls in an echocardiogram. Thus,

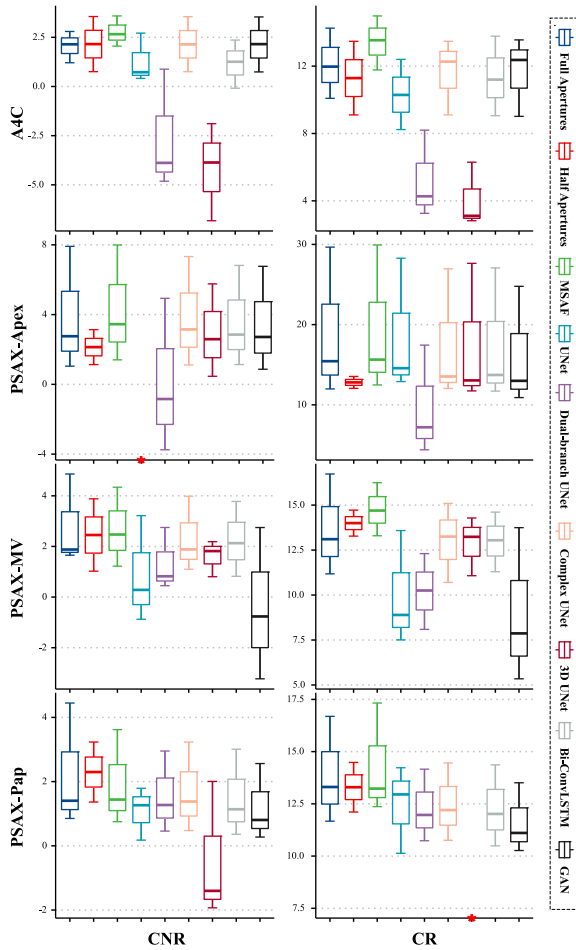


Fig. 6. The box plots of contrast analysis, visualizing CNR and CR for each benchmark model on four standard echocardiographic views. The red $\star$ on the horizontal axis denotes the method excluded from contrast analysis due to its poor image quality.

the 3-rd and 8-th difference maps showed that D_a constrained the signal quality in the heart walls.

E. Zero-Shot Generalization Assessment

Fig. 9 shows the results on the independent testing sets of unseen A2C and PLAX views. The MSAF achieved PSNRs of 32.57 (30.38) and 32.61 (30.36) dB on the recovered IQ signals (B-mode images) in the two imaging views, along with an SSIM of 0.87 (0.82) for both views. Despite an average performance decline of 4.93% on the zero-shot testing set compared to the trained views, MSAF remained superior to the nine benchmark models, which suffered from larger performance drops of 5.31%, 6.27%, 7.46%, 6.01%, 8.49%, 9.15%, 9.92%, 2.13%, and 9.21%, respectively. Additionally, MSAF trained with cardiac data achieved PSNR of 32.22 (29.89) dB and SSIM of 0.87 (0.82) on abdominal ultrasound IQ signals (B-mode images). Fig. 10 shows B-mode images of A2C view, PLAX view, and abdominal views recovered by MSAF. The results

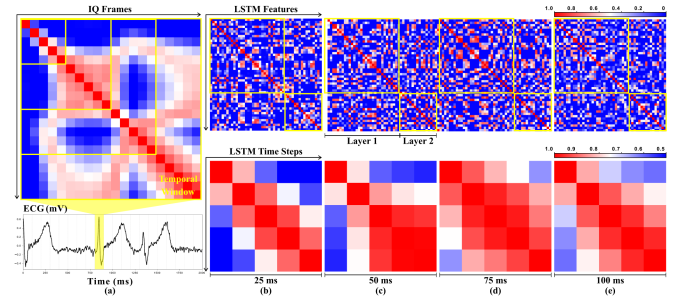


Fig. 7. (a) The cosine similarity between 20 IQ frames within a QRS complex in the ECG signal (yellow shaded segment); yellow boxes indicate four temporal windows: 25 ms, 50 ms, 75 ms, and 100 ms. (b)–(e) show the heat maps of cosine similarity between learned LSTM's features (top row) and between LSTM's time steps (bottom row) in the four cases of temporal windows.

demonstrated MSAF's capability of adapting to new imaging geometries and organs.

F. Health-Compromised Generalization Assessment

Fig. 11 shows the results on the independent testing sets with chronic conditions. MSAF achieved PSNR values of 33.49 (30.93), 33.47 (30.85), 33.49 (30.71), and 32.80 (30.10) dB on recovered IQ signals (B-mode images) in the PSAX-MV, PSAX-Pap, PSAX-Apex, and A4C views, respectively. The SSIM values for these views were 0.88 (0.84), 0.87 (0.83), 0.88 (0.83), and 0.87 (0.73), respectively. While MSAF exhibited an average performance reduction of 4.47% on chronic conditions compared to healthy samples, this performance decline was less severe than that observed in the other models, whose performance respectively dropped by 4.85%, 5.78%, 7.14%, 5.51%, 8.13%, 8.77%, 9.53%, 8.85%, and 8.51%. The results demonstrated the strong generalizability of MSAF in dealing with unseen health-compromised data.

G. Computational Load

Table II compares computational requirements across models, including the parameter sizes (# Params), floating point operations (FLOPs), inference time, and GPU memory consumption. They were tested on an NVIDIA RTX 3090 GPU (batch size = 1). The results show that MSAF's generator consumed 1,654MB of GPU memory and took 174.43 ms per inference for five frames. The three benchmark models that consumed less GPU memory than MSAF recovered only one frame per inference. Therefore, MSAF's balance between efficiency and throughput demonstrates its superior compatibility with resource-constrained clinical environments.

VI. DISCUSSION

This study proposed a multi-level sensing adversarial framework to recover the IQ signals in synthetic aperture ultrasound imaging. Extensive experiments were conducted to evaluate our method on a large dataset of in vivo human hearts (Fig. 4).

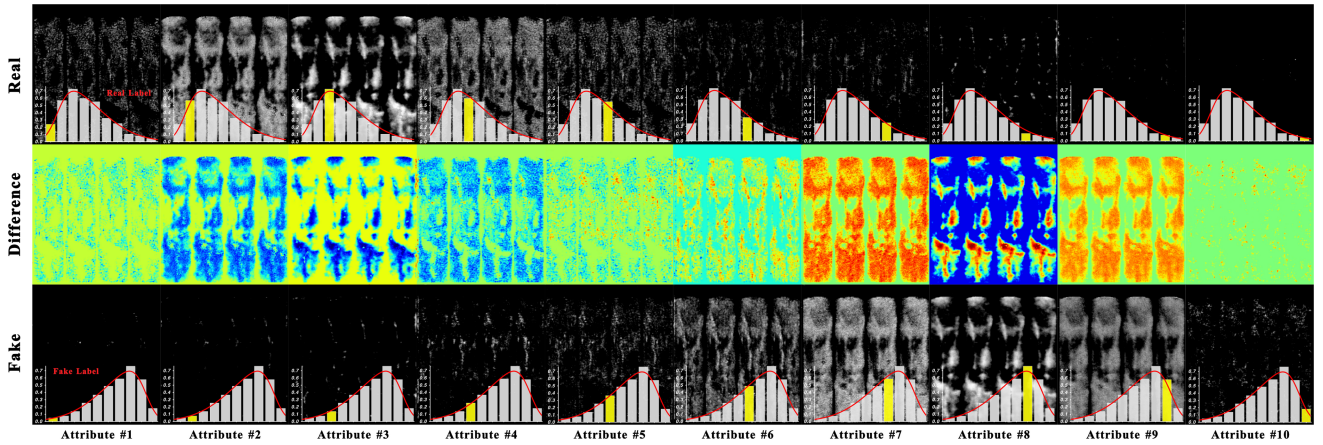


Fig. 8. Visualization of aperture-wise attribute maps. The upper and lower rows show attribute maps of the real (target GT) and fake (recovered by the generator) signals, respectively. The middle row shows the difference between real and fake attribute maps. The first to the 10-th attribute maps are shown from left to right, where four sub-apertures are placed horizontally in each attribute map. Red curves show the real or fake distribution labels. The histograms show the average values of 10 attribute maps, where the bin of the corresponding attribute is colored yellow.

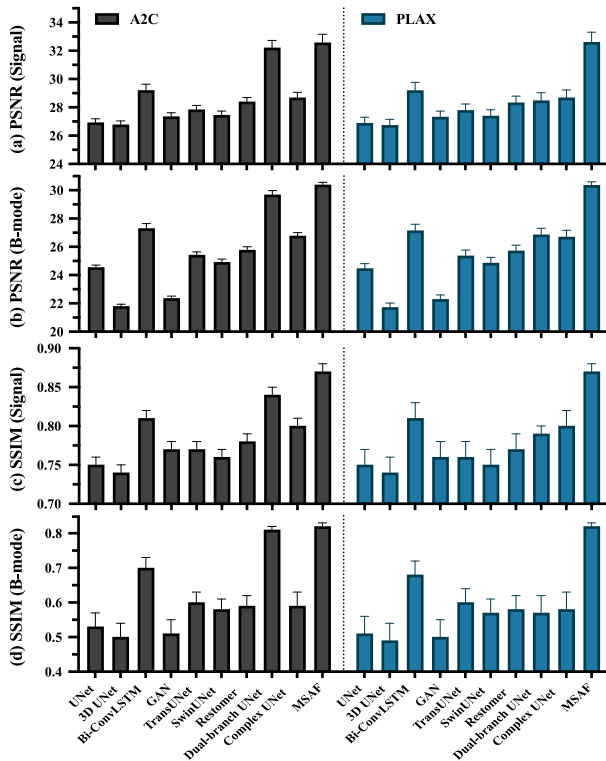


Fig. 9. The results on the zero-shot A2C and PLAX imaging views. (a), (b) The PSNR on the recovered IQ/RF signals and B-mode images. (c), (d) The SSIM on the recovered IQ/RF signals and B-mode images.

Qualitative and quantitative results, including PSNR and SSIM (Table III), errors (Fig. 5), as well as contrast (Fig. 6), demonstrated the superiority of MSAF in IQ signal recovery. The ablation studies (Table IV), visualization of the learned aperture-wise attributes (Fig. 8), and the temporal window analysis (Fig. 7) also verified the significance of the multi-level sensing strategy in the recovery process.

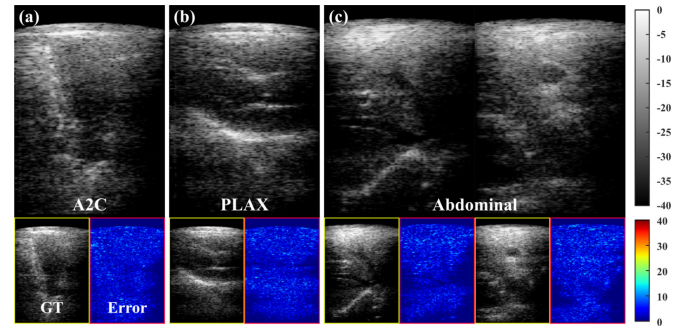


Fig. 10. B-mode images from one recovered sub-aperture by MSAF for (a) the heart in the A2C view, (b) the heart in the PLAX view, and (c) the liver in zero-shot testing.

Complex-valued neural networks are essential for MSAF to efficiently sense and recover IQ signals, as evidenced by ablation studies w/o G_u and G_l in Table III. This agreed with studies on complex signal processing [51] and the literature that Complex UNet [38] was more effective than real-valued networks when handling complex data. IQ signals also exhibit considerable advantages over RF signals in data volume and computational efficiency (Table II). By incorporating spatiotemporal and global-local sensing into complex neural networks, MSAF further improved signal recovery quality. This revealed that a great deal of valuable information remained hidden in the high-dimensional IQ signals and could be extracted by multi-level sensing strategies.

For sequential data processing, Figs. 4–6 show that ConvLSTM [41] outperformed 3D CNNs [40]. MSAF adopted bidirectional ConvLSTM as a temporal refinement module, and we modified ConvLSTM to a complex-valued neural network for IQ data adaptation. As demonstrated by the ablation study w/o G_l , the proposed complex Bi-ConvLSTM was instrumental because it improved temporal feature consistency.

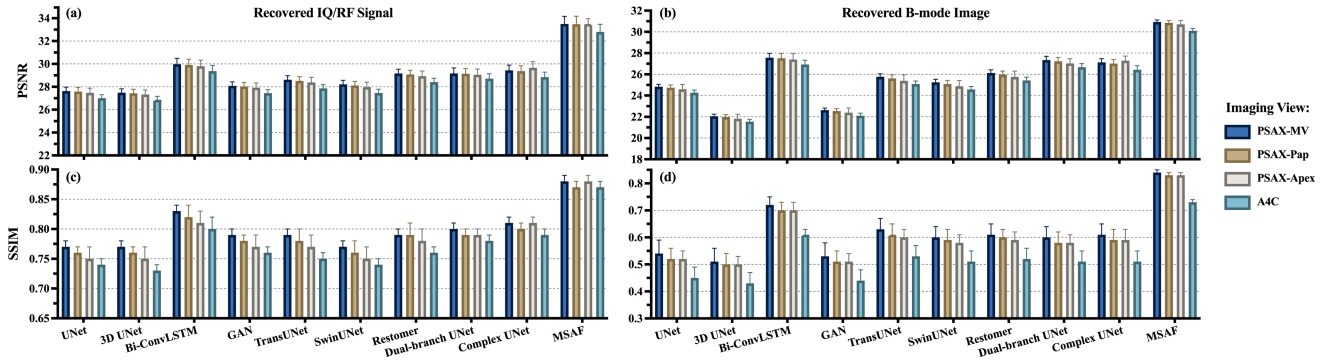


Fig. 11. Signal recovery performance of MSAF and nine benchmark models on the independent testing set with chronic conditions in four echocardiographic views. (a), (b) PSNRs and (c), (d) SSIMs for the recovered IQ/RF signals and B-mode images.

In addition to spatiotemporal sensing, MSAF encodes global-local sensing via an adversarial learning paradigm. In Fig. 4, MSAF led a global restoration performance closer to the real signals. Fig. 8 shows that MSAF also constrained local image attributes, such as anatomical details and speckle characteristics. Compared with standard GANs [34], MSAF simultaneously monitors global and local details by frame-wise global discrimination and aperture-wise attribute discrimination, both of which are the basis for further error reduction (GAN vs. MSAF in Fig. 5), contrast enhancement (GAN vs. MSAF in Fig. 6), PSNR and SSIM improvement (GAN vs. MSAF in Table III). MSAF does not directly impose global or local constraints on 2D images; instead, it considers the high dimensionality of synthetic aperture ultrasound signals. Specifically, frame-wise global discrimination performs an intra-frame but cross-aperture constraint. Aperture-wise attribute discrimination performs an intra-aperture but cross-frame constraint. The results of the ablation models w/o D_f and w/o D_a show that it is indispensable to design such a multi-level sensing strategy for high-quality signal recovery.

Numerous studies have investigated ultrasound IQ signals in either Cartesian or polar form [18], [19], [22], [23], [24], [39]. There is still an ongoing debate on which domain is better suited to recover the signal. Compared with other studies that make constraints in a single domain, MSAF introduces a domain converter into an adversarial learning process to supervise the recovered signals in both Cartesian and polar forms. This straightforward yet effective design for enhancing the recovery quality was validated by ablation study w/o domain converter.

A. Limitations

We currently use equally spaced transmit sub-apertures to show the feasibility of sparse aperture signal recovery. More flexible or random sparse transmission schemes will be explored in future work. Table II shows that MSAF required the highest FLOPs among all the models evaluated. The integration of Bi-ConvLSTM boosted MSAF's temporal-sensing ability but imposed a heavy FLOPs consumption. A more lightweight MSAF is warranted in the future to improve computational efficiency.

VII. CONCLUSION

This paper proposes a multi-level sensing adversarial framework tailored for IQ signal recovery in synthetic aperture ultrasound with sparse apertures. The framework incorporates spatiotemporal and global-local sensing. This approach effectively models the rich high-dimensional characteristics of synthetic aperture ultrasound signals and preserves their intrinsic correlations throughout the recovery process. Evaluations conducted on large in vivo human datasets substantiated MSAF's superiority in recovering high-fidelity, high-contrast, detail-rich, and temporally stable IQ signals with 32.80 dB PSNR, 0.88 SSIM, 2.92 dB CNR, and 15.46 dB CR. Compared with benchmark models, MSAF obtained average improvements of 13.28% and 7.62% on PSNR and SSIM, respectively. When generalized to zero-shot imaging targets and pathological data, MSAF continued to achieve the best performance, demonstrating strong robustness. The proposed framework creates new possibilities for complex-valued signal recovery in high-frame-rate and volumetric ultrasound with low hardware complexity.

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